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A Contemporary Research on Perceptual Expert System for Medical Diagnosis through Neural Dust Sensor

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Abstract

In the past few years many researches have been dedicated towards Neural dust sensor such as Deep Brain Simulation (DBS), Brain Machine interface (BMI), Ultrasonic backscattering system and so on. Mainly the Brain-Machine Interface is considered as the neural interface system that can be usable for a lifetime and to increase the life span of the sensor with the implementation of wireless electromagnetic neural interface. By implementing all this to detect the blood clot in human brain and to reduce the blood clot using ultrasonic waves and further other implementation in medical field. And, Neural Dust sensor is considered as the central nervous system of a particular system. This study aims to provide a review on usage of neural dust sensor in medical diagnosis.

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1. Introduction

In the last 5-8 years' various technologies seek new solutions for improving and monitoring the nervous system treatment techniques^{1,2,3,4,5}. For example, many solutions for the treatment have been developed such as placing

electrodes in the brain and to monitor the disease. The main drawback of this type of treatment is that insertion of electrodes and cables in brain may cause other problems also dangerous which are not practical everyday use for the patients. In this paper, the implementation of the ultrasonic sound ways through neural dust sensor to reduce or removal of the blood clot has been proposed^{6,7,8,9}. As we know that neural dust sensor only used as the monitoring system like receiving back the signal towards transverse through backscattering. This neural dust sensor should be thousand of 10-100 μm scale; the ultrasonic power has nearly 7% power efficiency links which is nearly -11.6 dB which further results in approximately 500 μW with 1 mm^2 ^{10,11,12,13}. The electromagnetic scale is also similar scale like sensor node scale of a neural dust sensor. Recently the majority of neural recording is done by potential change by depolarization effect and they are also called as action potentials which are also useful in dealing with diabetes neuro simulation and many other problems related to nervous systems. Neural dust is nothing but a sensor which is wireless and which is millimeter sized sensor^{14,15,16,17}. It was first discovered by the University of California, Berkeley. This sensor is also called as the nerve sensor. Fig. 1 shows the pictorial of the neural sensor.

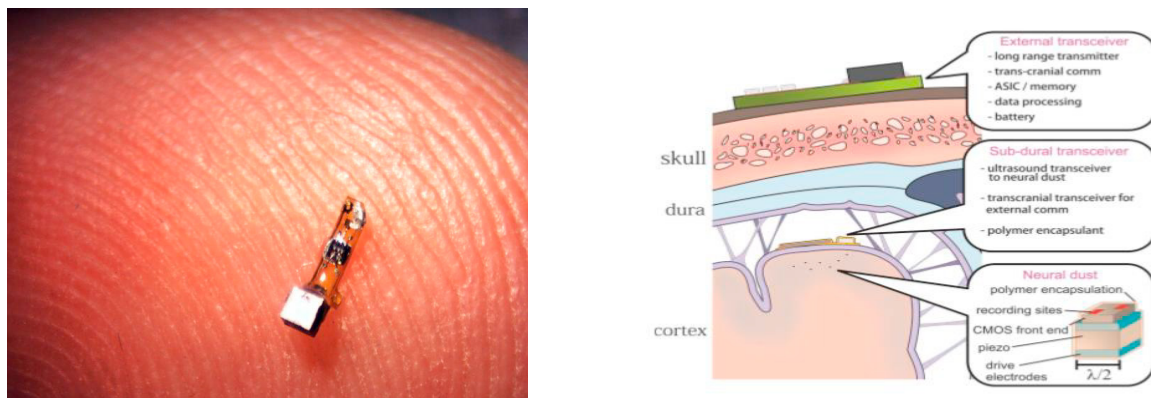


Fig. 1: (a) Neural Dust sensor; (b) Brain-machine interface representations showing the ultrasonic sensor with neural dust

2. Brain Machine Interface:

As the above Fig. 1, which represents how the neural dust sensor is placed in brain and how it collects the data, receive and transmits the data accordingly with the help of sub-Dural transceiver, external transceiver and neural dust. As above diagram the neural dust sensor is an electromagnetic transformer with ϕ turn ratio with the length of $\lambda/2$. At some frequency change the ultrasonic sound wave under go about 180° shift respectively.

3. Deep Brain Simulation

Deep brain simulation is nothing but similar to the brain machine interface where neural dust sensor is placed inside the brain for monitoring purpose and also deep brain simulation is somewhat different from neural dust sensor as it contains LED light which interface with neurons in brain for simulation purposes^{18,19,20,21}. This deep brain simulation is also helpful to treat the conditions like Dystonia, Epilepsy, Essential tremor, Parkinson's disease. Not only it can treat the conditions like this but also some of the potential treatments of deep brain simulation are Addiction, Dementia, Depression, Stroke Recovery, and Cluster Headache. As mentioned in the below Fig. 2, shows the interfacing of brain with ultrasound transducer and WI-opt neural dust^{22,23,24}.

4. Architecture of WI-neural dust

The Fig. 3 represents how the circuits are formed along with its working of piezoelectric nanowires with the help of energy harvester and rectifier. Thus, how the switching connections are placed and hoe the LED light works for interfacing and simulation of neurons. Ultrasonic sound waves help to vibrate the piezoelectric nanowires and by doing so the mechanical energy converts into electrical energy^{25,26,27}. As we know that the safety limit of 720mW/cm². Thus, the electric energy powers the LED light which produces the ultrasonic sound waves which

helps in implementing the monitoring of the system which is very sensitive for the specific frequency of the sensor²⁸.

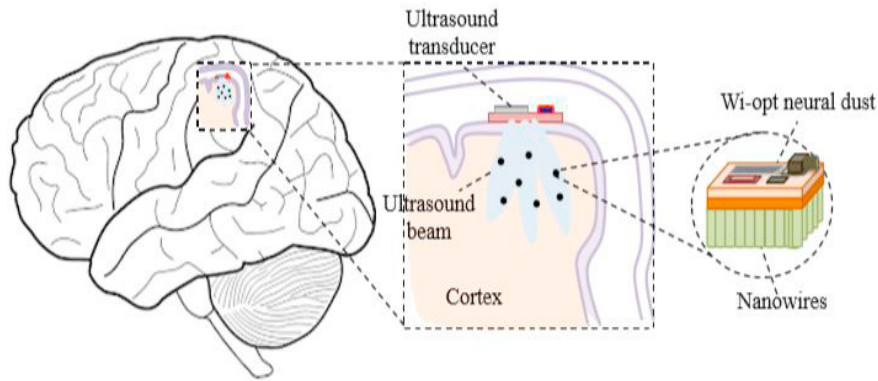


Fig. 2: Representation of deep brain simulation with WI-opt neural dust sensor

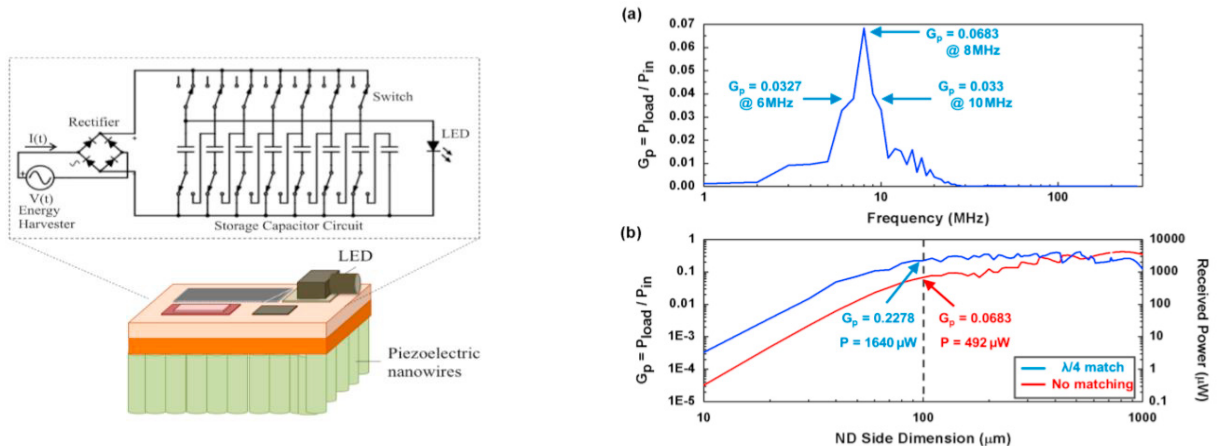


Fig. 3: (a) WI-neural dust sensor architecture; (b) Graphical representation of frequency Vs ND side Dimension

5. Ultrasonic Power Efficiency Vs Operating Frequency

The Fig. 3 shows the graphical representation of the frequency vs Neural dust side dimension (b) and the Ultrasonic power efficiency vs Operating frequency (a) here in the above graph shows clearly that electrical activity of neurons is most commonly measured by electrical potential across the cellular membrane^{29,30,31,32,33,34}. Anyhow this method is not so widely useful because of the drawbacks such as implantable recordings, electrical activity, and the measurements which are taken outside the membrane cells^{35,36,37,38,39,40}.

6. Test for Neural Dust Sensor for a Rat

Thus, the above Fig. 4 represents how the dust sensor is placed in leg of a rat and is interfaced with PC programming, which also helps in data transferring such as transmitting and receiving the signal obtained with the help of the neural dust sensor^{41,42,43,44,45}. The time Vs voltage represents the graphical representation of the reference input with the dust sensor output. Thus, this test is not only for the rat but also for different living species to monitor the functions of nervous system of a required system^{46,47,48,49,50}.

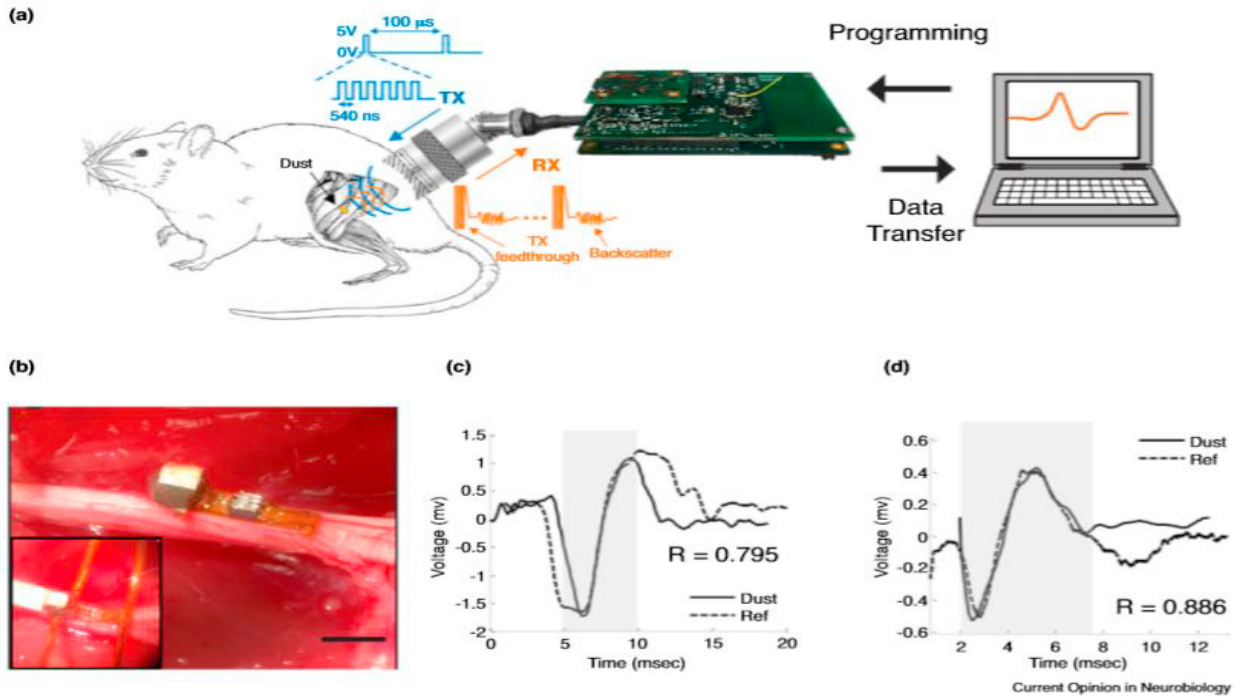


Fig. 4: Representation of working of a neural dust sensor on a rat

7. Simulated Results of Sensitivity

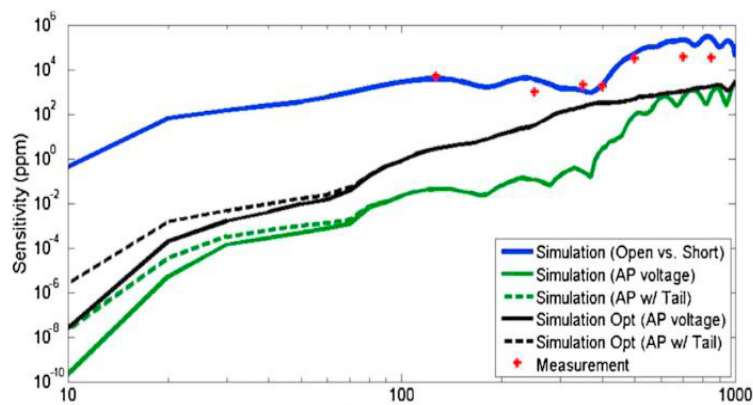


Fig. 5: Backscattering sensitivity

Here the Fig. 5 represents the Backscattering sensitivity of neural dust sensor with different variations such as for the simulation of neural sensor with AP voltage, open vs short simulation, Simulation opt (AP voltage and also with Tail) and also the manually measured values mathematically).

8. Future Scope

- The Neural Dust Sensor in future may use as a control of the central nervous system with simple interface with the brain^{51,52,53,54}.
- It also helps to remove the tumors present inside the brain and also on the neurons^{55,56}.

- Parkinson's is also a disease similar to brain disease which causes due to low dopamine, the symptoms shown are shakes in the limbs for slowing down the voluntary movement. Thus, this can be treated with the help of Neuron Dust Sensor⁵⁷.
- Alzheimer's is caused due to low acetylcholine; the symptoms of this disease are memory loss. This disease can be implemented by implementing the Neural Dust Sensor in Brain Nervous system^{58,59,60}.

9. Conclusion

Thus, most of the nervous breakdown issues are treated by Brain Machine Interface, Deep Brain Stimulation, ultrasonic wireless optical sensors, etc. the main disadvantages of this treatment methods are that they are not useable for the lifetime usage. By implementing ultrasonic sound waves in the neural dust sensor, we can reduce or complete removal of the blood clot present inside the brain as well as it is used to monitor and detects the tumors in advance and it gives warning in the form of a transmission signal. Thus, by following the implementations of the Neural Dust Sensor it helps to cure most of the diseases related to the brain and nervous system such as Schizophrenia, Parkinson's, Alzheimer's, Brain tumors, etc.

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